MATH317 · NUMERICAL METHODS

Chapter 3.2 Newton's Method

Midterm summary · Lecture 6

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What's inside

01 What is Newton's method?

02 Where does the formula come from?

03 The slide example

04 An example showing Newton's speed

05 Convergence analysis

06 When do we stop?

07 Newton vs Bisection

08 How it shows up in the midterm

09 Common mistakes

10 Check yourself

01 What is Newton's method?

A method for finding a root of `f(x) = 0`. It is also called the **Newton-Raphson iteration**. It is one of the most important procedures in numerical analysis and is much faster than bisection.

Requirement

The function must be **Differentiable**, because the method needs `f'(x)`.

The formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Start from a guess x_0 and substitute to get x_1. Substitute x_1 to get x_2, and so on.

02 Where does the formula come from?

First view: the tangent line

At the point $(x_0, f(x_0))$ draw the tangent. Near that point the tangent is close to the curve, and its equation is:

$$l(x) = f'(x_0)(x - x_0) + f(x_0)$$

Instead of the root of the hard function, find the root of the tangent. Set $l(x) = 0$ and solve:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Second view: Taylor series

Ask: what correction h added to x_0 gives the root? We want $f(x_0 + h) = 0$.

$$f(x_0 + h) = f(x_0) + h\, f'(x_0) + \frac{h^2}{2!} f''(x_0) + \ldots = 0$$

Keep the first two terms only: $f(x_0) + h\, f'(x_0) = 0$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Both views give the same formula.

03 The slide example

If $f(x) = x^3 - x + 1$ and $x_0 = 1$, what are x_1 and x_2 in the Newton iteration?

1 Differentiate

$f(x) = x^3 - x + 1$ $\qquad$ $f'(x) = 3x^2 - 1$

2 Compute x_1

$f(1) = 1 - 1 + 1 = 1$ $\qquad$ $f'(1) = 3 - 1 = 2$

$$x_1 = 1 - \frac{1}{2} = \frac{1}{2}$$

3 Compute x_2

$$f\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{1}{2} + 1 = \frac{5}{8} \qquad f'\left(\frac{1}{2}\right) = \frac{3}{4} - 1 = -\frac{1}{4}$$

$$x_2 = \frac{1}{2} - \frac{\frac{5}{8}}{-\frac{1}{4}} = \frac{1}{2} + \frac{5}{2}$$

$$x_2 = 3$$

Notice

Here $x_2 = 3$ jumped far from x_0. Newton's method does not guarantee convergence when the first guess is far from the root.

04 An example showing Newton's speed

The slides solve $x^3 + x = 2x^2 + 3$. Move everything to one side:

$$f(x) = x^3 - 2x^2 + x - 3 = ((x - 2)x + 1)x - 3$$

$$f'(x) = 3x^2 - 4x + 1 = (3x - 4)x + 1$$

Both are written in **Nested form** so they are cheaper to evaluate. Start from $x_0 = 3$:

n	x_n	$f(x_n)$
0	3.0	9.0
1	2.4375	2.04
2	2.21303	0.256
3	2.17555	6.46×10^{-3}
4	2.17456	4.48×10^{-6}
5	2.17456	2.2×10^{-12}

The real root is `2.17456`. Watch how fast `f(xₙ)` shrinks: the number of correct digits roughly doubles every step.

05 Convergence analysis

Let the error at step n be `eₙ = xₙ − r`, where `r` is the root.

Theorem 1

The errors of Newton's method are related by:

$$e_{n+1} = \tfrac{1}{2} \cdot \frac{f''(\xi_n)}{f'(x_n)} \cdot e_n^2 \qquad (\xi_n \text{ between } x_n \text{ and } r)$$

$$|\, e_{n+1} \,| \;\le\; c \cdot |\, e_n \,|^2$$

The new error is proportional to the square of the old one. This is called **Quadratic convergence**.

If the error is now	After one step it is about
10^{-2}	10^{-4}
10^{-4}	10^{-8}

Condition

This holds only when x_0 is sufficiently close to the root.

06 When do we stop?

After each step compute the **Absolute relative approximate error**:

$$|\varepsilon_a| = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| \times 100$$

1 **Compute $|\varepsilon_a|$.**

2 **Compare it with the given tolerance ε.**

3 **If $|\varepsilon_a| > \varepsilon$ continue. Otherwise stop.**

Also stop when the number of iterations exceeds the allowed maximum.

The pseudocode on the slide

It starts from a value `x` and runs at most `n_max` iterations. It needs two functions, `f(x)` and `f'(x)`. The parameters ε and δ control convergence and accuracy.

07 Newton vs Bisection

	Bisection	Newton
What it needs to start	An interval `[a, b]` with a sign change	One guess x_0
Needs the derivative?	No	Yes
Speed	Slow: the error halves each step	Fast: **Quadratic convergence**
Is convergence guaranteed?	Yes	No, unless x_0 is close to the root
Work per step	One evaluation: `f(c)`	Two evaluations: `f` and `f'`

08 How it shows up in the midterm

Newton's method appeared in one of the four midterms (461), for five marks.

> Given $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, apply Newton's method for two iterations.

$$f'(x) = 3x^2 - 2$$

$$f(1.5) = 3.375 - 3 - 1 = -0.625 \qquad f'(1.5) = 6.75 - 2 = 4.75$$

$$x_1 = 1.5 - \frac{-0.625}{4.75} = 1.5 + 0.1316$$

$$\mathbf{x_1 = 1.6316}$$

$$f(1.6316) = 0.0803 \qquad f'(1.6316) = 5.9864$$

$$x_2 = 1.6316 - \frac{0.0803}{5.9864} = 1.6316 - 0.0134$$

$$\mathbf{x_2 = 1.6182}$$

If the question also asks for the error between the last two values:

$$|\varepsilon_a| = \left|\frac{1.6182 - 1.6316}{1.6182}\right| \times 100 = 0.83\ \%$$

09 Common mistakes

Mistake	Fix
Adding instead of subtracting in the formula	$x_{n+1} = x_n - \frac{f}{f'}$. Watch the sign when f is negative.
Differentiating incorrectly	Write $f'(x)$ on the first line and check it.
Substituting x_0 again in the second step	The second step uses x_1.
Rounding x_1 too much before continuing	Keep at least four decimal places.
Saying Newton always converges	No. It needs x_0 close to the root.

10 Check yourself

1. $f(x) = x^2 - 2$ and $x_0 = 1$. Find x_1 and x_2.
2. $f(x) = x^3 - x - 1$ and $x_0 = 1.5$. Find x_1 and x_2.
3. What does quadratic convergence mean?

Solutions

1) $f'(x) = 2x$

$$x_1 = 1 - \frac{1 - 2}{2} = 1 + 0.5 = 1.5$$

$$x_2 = 1.5 - \frac{2.25 - 2}{3} = 1.5 - 0.0833$$

$\mathbf{x_2 = 1.4167}$ (true root $\sqrt{2} = 1.4142$)

2) $f'(x) = 3x^2 - 1$

$f(1.5) = 0.875$ $f'(1.5) = 5.75$

$$x_1 = 1.5 - \frac{0.875}{5.75} = 1.3478$$

$f(1.3478) = 0.1007$ $f'(1.3478) = 4.4499$

$$\mathbf{x_2 = 1.3478 - \frac{0.1007}{4.4499} = 1.3252}$$

3) The new error is proportional to the square of the old one: $|e_{n+1}| \le c\,|e_n|^2$. The number of correct digits roughly doubles each step.